

REQUIREMENT GATHERING

DATE: 21-07-2026

CampusPulse- A Semantic Recruitment Intelligence & Campus Placement Automation System

1. Project Overview

CampusPulse is a full-stack, AI-enhanced placement automation system designed to eliminate the manual chaos, scheduling conflicts, and communication gaps that plague campus recruitment drives. Currently, placement activities rely on fragmented spreadsheets, endless email chains, and verbal confirmations—leading to double-bookings, ineligible student applications, missed communications, and a lack of transparency in candidate selection.

Main Objectives:

- **Automate the entire recruitment lifecycle**—from job drive creation to final offer acceptance.
- **Introduce semantic intelligence** using Sentence-BERT to match candidates based on contextual resume understanding rather than superficial keyword matching.
- **Ensure transparency and fairness** through an Explainable AI (XAI) module that shows students *why* they were shortlisted or rejected.
- **Eliminate double-bookings** across aptitude tests, group discussions, and interviews via a real-time conflict-detection algorithm.
- **Support multi-stage recruitment pipelines**—recruiters can configure aptitude tests and group discussions before interviews.
- **Close the recruitment loop** by providing a complete offer-to-acceptance lifecycle with secure offer letter delivery.

2. System Scope

The system is proposed as a **fully functional prototype** for a **single college** deployed on **localhost**.

Aspect	Scope
Deployment	Localhost (Node.js + Python + MongoDB running natively)
Users	Students, Recruiters, and TPOs of a single institution
Core Features	Authentication, Profile Management, Job Drive Management, Verification Pipeline, Multi-Stage Pipeline (Aptitude/GD/Interview), Semantic AI Ranking, Explainable AI (XAI) Feedback, Conflict-Detection Algorithm, Offer-to-Acceptance Lifecycle, CSV Bulk Import, Recruiter Trust Score, Automated Email Reminders, TPO Dashboard, PWA Offline Support
Out of Scope	Private Chat, Mass Broadcast, Microservices, Docker, CI/CD, Cloud Deployment, Multi-Tenancy

3. Target Audience

The system is designed for the following stakeholder groups:

Audience Group	Description
Final-Year Students	Actively seeking employment through campus placements. They need eligibility clarity, real-time status updates, interview schedules, and transparent shortlisting feedback.
Company Recruiters	HR representatives and hiring managers who post job drives, review applications, schedule tests/GDs/interviews, declare results, and upload offer letters.
Training & Placement Officers (TPOs)	College administrators responsible for overseeing the entire placement process, approving recruiters, bulk-importing student data, and monitoring placement analytics.
Academic Evaluators / Project Guides	Faculty members and reviewers who assess the project's functionality, design, and documentation.

4. Modules

The system consists of the following key modules:

SL.NO	Module Name	Functionality
1	Authentication & Authorization	Secure JWT-based login with role-based access control (Student, Recruiter, TPO).
2	Profile Management	Students upload resumes (PDF); Recruiters submit company profiles; TPOs manage approvals.
3	Job Drive Management	Recruiters create job postings with CTC, deadlines, CGPA/branch/backlog filters.
4	Verification Pipeline	Auto-blocks ineligible students (based on CGPA, branch, backlogs) before they can apply.
5	Multi-Stage Recruitment Pipeline	Recruiters configure Aptitude Tests and Group Discussions as optional stages before interviews. Students progress through: Applied → Aptitude Scheduled → Aptitude Completed → GD Scheduled → GD Completed → Interview Scheduled → Interview Completed → Selected/Rejected/Waitlisted → Offer Accepted/Declined → Placed.
6	Semantic Resume Intelligence	Python microservice using Sentence-BERT to generate contextual embeddings and rank candidates based on meaning rather than keywords.
7	Explainable AI (XAI) Feedback	Displays specific sentences from the student's resume that influenced their shortlisting or rejection score.
8	Conflict-Detection Algorithm	MongoDB aggregation pipeline that validates proposed slots for tests, GDs, and interviews against existing commitments, blocking overlaps in real-time.
9	Scheduling	Recruiters assign specific time slots for Aptitude Tests, Group Discussions, and Interviews to shortlisted candidates.
10	Recruiter Result Declaration	Recruiters mark candidates as Selected/Rejected/Waitlisted after interviews.
11	Offer Letter Management	Recruiters upload official offer letters (PDF); Students view/download and formally Accept/Decline the offer.

12	CSV Bulk Import	TPOs upload Excel/CSV files to pre-create student accounts in bulk.
13	Recruiter Trust Score	Automated verification using WHOIS domain age and MCA/ROC data to assist TPOs in approving legitimate recruiters.
14	Automated Email Reminders	node-cron scheduler sends 24-hour reminders for tests, GDs, interviews, and result declarations via Nodemailer.
15	TPO Analytics Dashboard	ECharts-based visualizations showing placed vs. unplaced students, branch-wise statistics, stage-wise funnel visualization, offer-to-acceptance ratios, and drop-off analytics.
16	Progressive Web App (PWA)	Service Workers and Web App Manifest enable offline access to schedules and app-like installation on mobile devices.
17	Live System Alerts	Socket.IO -powered pop-up notifications when recruiters schedule tests/GDs/interviews or update statuses.

5. User Roles

The system defines three distinct user roles with granular permissions:

User Role	Who They Are	Access Level / Permissions
Student	Final-year undergraduate/postgraduate students.	View eligible drives, apply, track multi-stage application status, view XAI feedback, receive live alerts, view/download offer letters, accept/decline offers, view schedules, receive email reminders, access PWA offline.

Recruiter	HR representatives and hiring managers from companies.	Register company (pending TPO approval), post job drives, configure multi-stage pipelines, view AI-ranked applications with XAI justifications, shortlist/reject applicants, schedule tests/GDs/interviews (conflict-checked), declare results, upload offer letters (PDF), track offer acceptance rates.
TPO	Training & Placement Officers of the institution.	Approve/reject recruiter registrations, bulk-import student data via CSV, view analytics dashboard (placed/unplaced, stage-wise funnel, offer-acceptance trends, drop-off analytics), oversee all active drives, override scheduling conflicts if necessary.

6. System Ownership

Aspect	Details
Primary Owner	The academic institution / student developing the system.
Target End-User	The institution's Training & Placement Office (TPO), final-year students, and recruiting companies.
Intellectual Property	The project is developed as an academic research contribution for the capstone project.
Hosting	Localhost for Semester 1; cloud deployment planned for future.

7. Industry / Domain

CampusPulse operates at the intersection of multiple domains:

Domain	Relevance
Education Technology (EdTech)	The platform digitizes and automates a core administrative function of higher education institutions—campus placements.
Human Resource Technology (HR-Tech)	It facilitates recruitment, candidate shortlisting, interview scheduling, and offer management—core HR functions.
Artificial Intelligence & Machine Learning	The system employs Sentence-BERT (NLP) for semantic resume matching and Explainable AI (XAI) for transparent shortlisting decisions.
Real-Time Communication	Socket.IO powers live system alerts.
Progressive Web Applications	PWA support provides offline accessibility for students.

8. Data Collection Contacts

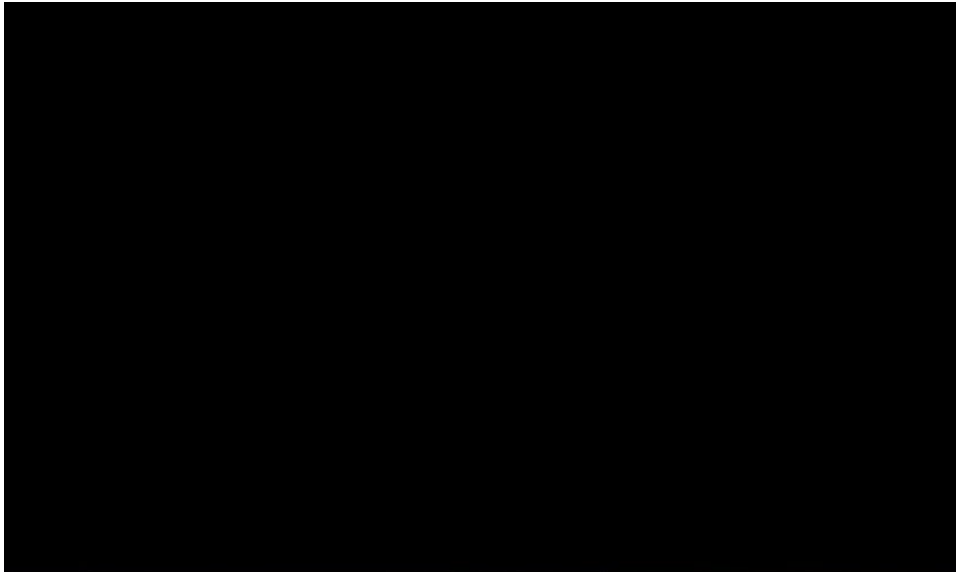
Contact Person:

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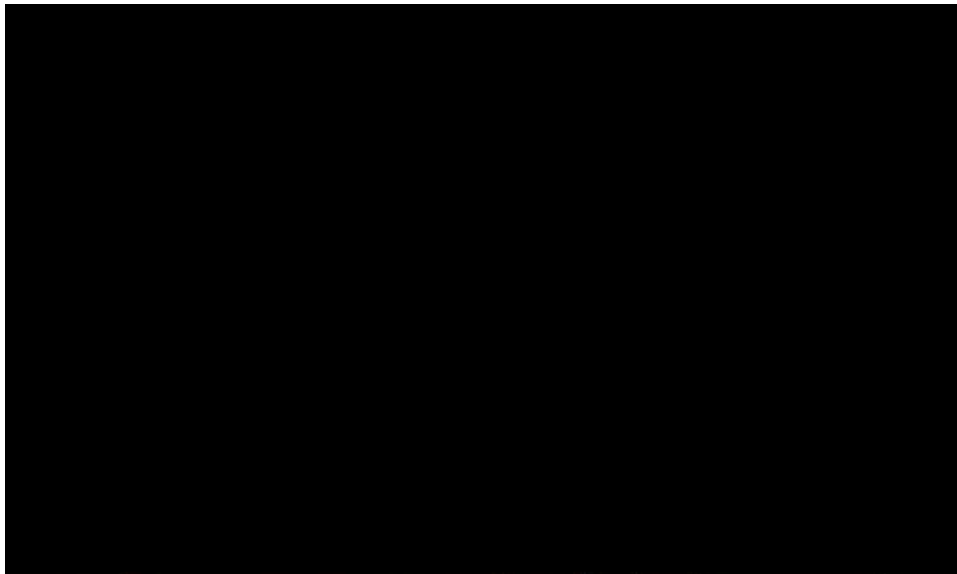
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9. Data Collection Questionnaire

1. Can you walk me through the entire placement process?

Companies reach out to us via email or phone. We share their JD with students via WhatsApp groups and notice boards. Students apply by filling a Google Form and submitting resumes to our office. We collect all resumes, filter them manually based on CGPA/branch, and share the shortlist with the recruiter. The recruiter visits campus conducts tests/GDs/interviews and shares the final selected list with us. We then collect offer letters and manually track who accepted.

2. What are the biggest frustrations or headaches in managing placements?

Two things keep me up at night. First, scheduling interviews—students often get double-booked and resolving that is a nightmare. Second, students constantly calling and emailing me for status updates. I spend more time answering queries than managing the process.

3. How do you currently make sure that only eligible students apply for a drive?

We use an ATS (Applicant Tracking System) to verify resumes. It scans the resumes for keywords and assigns a score based on how well they match the job description. But it's not perfect—it still doesn't filter students at the application stage. Students can still apply even if their score is low, and we manually have to shortlist the final list.

4. How do you communicate important updates to students and recruiters?

We use WhatsApp groups and mass emails.

5. What is the most common complaint you hear from students about placements?

Nothing. Students don't really complain to me directly. They usually go to their class representatives or faculty advisors. The feedback I get is filtered through them. I don't have specific complaints that students have shared with me personally

6. What feedback do recruiters usually give you about the current process?

Recruiters generally provide their feedback through a feedback platform available at the placement office. They are usually comfortable with the current process and don't complain much. However, they do appreciate it when we make things smoother for them.

7. How do you currently manage offer letters and final placement records?

We collect offer letters via email or physical copies. We maintain an Excel sheet to track which students accepted or declined.

8. If an AI system could rank students based on their resumes, would you trust it?

I would trust it if I could see why a student got a particular rank. If the system just gives a number, I wouldn't use it. But if it shows me which parts of the resume matched or didn't match, then yes—that would save me and the recruiters a lot of time.

9. What do you think of the limitations in developing an automation system for placement and recruitment?

Automation is helpful, but it can't replace human judgment. AI can't judge soft skills or body language during interviews. Also, the system is only as good as the data we feed it—if students enter wrong CGPA, the results are useless. Getting everyone to use a new platform is tough. People are comfortable with WhatsApp and emails. And what if the server crashes during placement season? That would be a disaster. Finally, placements involve human relationships—trust with recruiters, counseling anxious students. Automation can help, but it can never fully replace the human touch.

10. Is there anything else you would like to share?

I wish students could track their own status instead of calling me every day. Also, if recruiters could directly update results and upload offer letters on the platform, it would save me from playing the middleman. Overall, anything that reduces manual work and improves transparency is welcome.